

Mathematics A

Unit 2 • Chapter OL-T38 Classified

Cross-session • chapter-organised candidate

16 classified items • 80 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 2 • Unit 2 • Chapter OL-T38 • 16 items • 80 marks

Chapter	Items	Marks	Starts
01. OL-T38 Differentiation	16	80	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Differentiation

Unit 2 • 80 marks • 16 item(s)

June 2025 | 4MA1-2H Q17 | 6 marks

Coordinates of polynomial stationary points

June 2024 | 4MA1-1H Q18 | 5 marks

Points where gradient equals a value

November 2023 | 4MA1-1H Q16 | 4 marks

Points where gradient equals a value

November 2023 | 4MA1-2H Q18 | 4 marks

Velocity and acceleration from displacement

June 2023 | 4MA1-1H Q11 | 6 marks

Points where gradient equals a value

November 2024 | 4MA1-1H Q20 | 4 marks

Equation of a calculus tangent

June 2022 | 4MA1-1H Q17 | 5 marks

Instantaneous rest or direction reversal

January 2022 | 4MA1-1H Q24 | 6 marks

Recover curve parameters from point and gradient data

January 2022 | 4MA1-2H Q24 | 5 marks

Instantaneous rest or direction reversal

January 2020 | 4MA1-1H Q15 | 5 marks

Maximum volume from geometric dimensions

January 2021 | 4MA1-1H Q17 | 5 marks

Maximum volume from geometric dimensions

January 2021 | 4MA1-2H Q19 | 5 marks

Instantaneous rest or direction reversal

June 2021 | 4MA1-2H Q21 | 5 marks

Recover curve parameters from point and gradient data

May-Nov 2020 | 1H Q23 | 4 marks

Points where gradient equals a value

January 2023 | 1H Q20 | 5 marks

Maximum volume from geometric dimensions

... 1 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

17 $y = 4x^3 + 5x^2 + 2x$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots (2)$$

(b) Find the coordinates of the turning points on the graph with equation $y = 4x^3 + 5x^2 + 2x$
Show clear algebraic working.

$$\dots\dots\dots (4)$$

(Total for Question 17 is 6 marks)

Worked Solution 28

Derivative and turning points

UNIT 2
6 marks**Differentiate**

$$\frac{dy}{dx} = 12x^2 + 10x + 2.$$

Solve derivative equal to zero

$$12x^2 + 10x + 2 = 2(3x + 1)(2x + 1) = 0 \Rightarrow x = -\frac{1}{3}, -\frac{1}{2}.$$

Substitute

$$x = -\frac{1}{2} \Rightarrow y = -\frac{1}{4}; \quad x = -\frac{1}{3} \Rightarrow y = -\frac{7}{27}.$$

Final answer

$$\frac{dy}{dx} = 12x^2 + 10x + 2; \quad \left(-\frac{1}{2}, -\frac{1}{4}\right), \left(-\frac{1}{3}, -\frac{7}{27}\right)$$

Question 08

4MA1-1H Q18

ORIGINAL QUESTION

5 marks

18 A curve C has equation $y = x^3 - 40x + 1$

Find the coordinates of both the points on C at which the gradient is 8

(..... ,)

(..... ,)

Worked Solution 08

Gradient on a curve

MODULAR UNIT 2

5 marks

Differentiate

$$dy/dx = 3x^2 - 40.$$

Set the gradient

$$3x^2 - 40 = 8 \Rightarrow x = \pm 4.$$

Coordinates

$$y(4) = -95, \quad y(-4) = 97.$$

Final answer

$$(4, -95), (-4, 97)$$

Question 13

4MA1-1H Q16 demand

ORIGINAL QUESTION UNIT 2

4 marks

16 A curve has equation $y = 4x^3 - 8x + 5$

Find the x coordinates of the two points on the curve where the gradient is $\frac{1}{3}$

.....

Worked solution 13

Coordinates with a given gradient

MODULAR UNIT 2

4 marks

Demand – Coordinates with a given gradient

Differentiate

$$\frac{dy}{dx} = 12x^2 - 8.$$

Set the gradient

$$12x^2 - 8 = \frac{1}{3} \Rightarrow 12x^2 = \frac{25}{3} \Rightarrow x^2 = \frac{25}{36}.$$

Take both roots

$$x = \pm \frac{5}{6}.$$

Final answer

$$x = \pm 5/6$$

Question 30

4MA1-2H Q18 whole

ORIGINAL QUESTION UNIT 2

4 marks

- 18 A particle is moving along a straight line that passes through the fixed point O
The displacement, s metres, of the particle from O at time t seconds is given by

$$s = 2t^3 - 5t^2 + 6t - 5$$

Find the value of t when the acceleration of the particle is 5 m/s^2

$t = \dots\dots\dots$

Worked solution 30

Acceleration from a displacement function

MODULAR UNIT 2

4 marks

whole – Acceleration from a displacement function

Differentiate once

$$v = \frac{ds}{dt} = 6t^2 - 10t + 6.$$

Differentiate again

$$a = \frac{dv}{dt} = 12t - 10.$$

Set the required acceleration

$$12t - 10 = 5 \Rightarrow 12t = 15.$$

Solve

$$t = \frac{15}{12} = 1.25 \text{ s.}$$

Final answer

$$t = 1.25 \text{ s}$$

Question 11

4MA1-1H Q11

ORIGINAL QUESTION

6 marks

11 The curve **C** has equation $y = 4x^3 + x^2 - 20x$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots$$

(2)

(b) Find the x coordinates of the points on **C** where the gradient is 4
Show clear algebraic working.

$$\dots\dots\dots$$

(4)

Worked Solution 11

Differentiate and solve a gradient condition

MODULAR UNIT 2

6 marks

Differentiate

$$\frac{dy}{dx} = 12x^2 + 2x - 20.$$

Set the gradient to 4

$$12x^2 + 2x - 20 = 4 \Rightarrow 6x^2 + x - 12 = 0.$$

Factorise

$$6x^2 + x - 12 = (3x - 4)(2x + 3).$$

Final answer

$$x = \frac{4}{3} \text{ or } x = -\frac{3}{2}$$

Original question 13

4MA1-1H Q20 M01

MODULAR UNIT 2

4 marks

20 The curve with equation $y = 2x^4 - 64x$ has a minimum point.

Find an equation of the tangent to the curve at the minimum point.

Show clear algebraic working.

.....

Worked solution 13

Find the tangent at a minimum

MODULAR UNIT 2

4 marks

M01 – Find the tangent at a minimum**Differentiate**

$$\frac{dy}{dx} = 8x^3 - 64.$$

Set the gradient to zero

$$8x^3 - 64 = 0 \Rightarrow x = 2.$$

Find the y-coordinate

$$y = 2(2^4) - 64(2) = -96.$$

State the horizontal tangent

$$y = -96.$$

Final answer

$$y = -96$$

Original question 15

4MA1-1H Q17 Q17

ORIGINAL QUESTION**5 marks**

- 17 A particle P moves along a straight line.
The fixed point O lies on this line.

The displacement of P from O at time t seconds, $t \geq 1$, is s metres where

$$s = 4t^2 + \frac{125}{t}$$

The velocity of P at time t seconds, $t \geq 1$, is v m/s

Work out the distance of P from O at the instant when $v = 0$

..... m

Worked solution 15

Chapter: Differentiation

MODULAR UNIT 2

5 marks

Q17 – Chapter: Differentiation

Family: Apply differentiation to linear kinematics • Variant: Instantaneous rest or direction reversal

Method

$$s = 4t^2 + 125t^{-1}, \text{ so } v = ds/dt.$$

*Why: Velocity is the derivative of displacement with respect to time.***Method**

$$v = 8t - 125t^{-2} = 8t - 125/t^2.$$

*Why: Differentiate each power of t .***Method**

$$\text{When } v = 0, 8t - 125/t^2 = 0, \text{ so } 8t^3 = 125 \text{ and } t = 2.5 \text{ seconds.}$$

*Why: Multiply by t^2 and solve the resulting cubic for $t \geq 1$.***Method**

$$\text{At this time, } s = 4(2.5)^2 + 125/2.5.$$

*Why: Substitute the stationary time into the original displacement expression.***Method**

$$s = 25 + 50 = 75, \text{ so the distance from O is 75 m.}$$

*Why: The displacement is positive, so its magnitude is the distance.***Final answer****distance: 75 m**

Original question 15

4MA1-1H Q24

ORIGINAL QUESTION**6 marks**

24 The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates $(2, -6)$ lies on **C**

The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3

Show clear algebraic working.

$y = \dots\dots\dots$

Worked solution 15

Chapter: Differentiation

MODULAR UNIT 2

6 marks

Q24 – Chapter: Differentiation

Family: Find and classify stationary points • Variant: Recover curve parameters from point and gradient data

Method

At $(2, -6)$, $-6 = 8a + 4b - 24 + 6$, so $8a + 4b = 12$.

Why: Substitute the point into the cubic equation.

Method

$\frac{dy}{dx} = 3ax^2 + 2bx - 12$.

Why: Differentiate the curve.

Method

At $x=2$, $16 = 12a + 4b - 12$, so $12a + 4b = 28$.

Why: Use the given gradient at A.

Method

Solving $8a + 4b = 12$ and $12a + 4b = 28$ gives $a = 4$ and $b = -5$.

Why: Solve the simultaneous linear equations.

Method

At $x=3$, $y = 4(3)^3 - 5(3)^2 - 12(3) + 6$.

Why: Substitute $x=3$ and the constants.

Method

$y = 33$.

Why: Evaluate the expression.

Final answer

y: 33

Original question 35

4MA1-2H Q24

ORIGINAL QUESTION**5 marks**

24 A particle P moves along a straight line that passes through the fixed point O

The displacement, x metres, of P from O at time t seconds, where $t \geq 0$, is given by

$$x = 4t^3 - 27t + 8$$

The direction of motion of P reverses when P is at the point A on the line.

The acceleration of P at the instant when P is at A is $a \text{ m/s}^2$

Find the value of a

$a = \dots\dots\dots$

Worked solution 35

Chapter: Differentiation

MODULAR UNIT 2

5 marks

Q24 – Chapter: Differentiation

Family: Apply differentiation to linear kinematics • Variant: Instantaneous rest or direction reversal

Method

$$v = dx/dt = 12t^2 - 27.$$

*Why: Differentiate $x = 4t^3 - 27t + 8$.***Method**At a reversal $v = 0$, so $12t^2 - 27 = 0$ and $t^2 = 9/4$.*Why: Set velocity equal to zero.***Method**Since $t \geq 0$, $t = 3/2$.*Why: Choose the physical time.***Method**

$$a = dv/dt = 24t.$$

*Why: Differentiate velocity.***Method**At $t = 3/2$, $a = 36 \text{ m/s}^2$.*Why: Evaluate the acceleration.***Final answer****acceleration:** 36 m/s^2

15

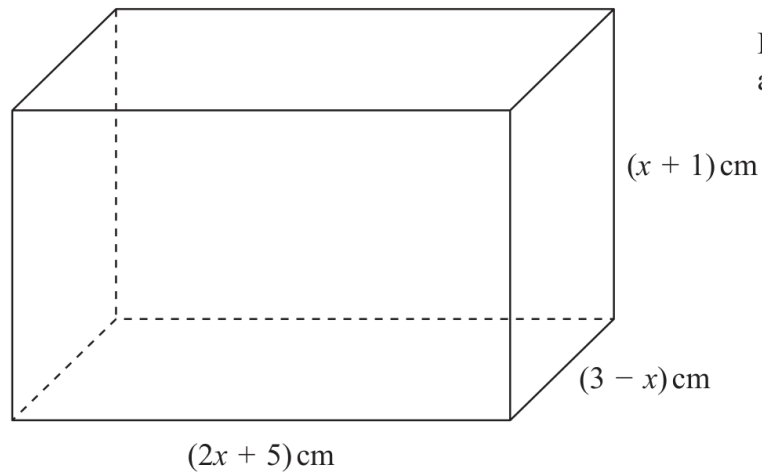


Diagram **NOT**
accurately drawn

The diagram shows a cuboid of volume $V\text{cm}^3$

(a) Show that $V = 15 + 16x - x^2 - 2x^3$

There is a value of x for which the volume of the cuboid is a maximum.

(b) Find this value of x .
Show your working clearly.
Give your answer correct to 3 significant figures.

$x = \dots\dots\dots$
(5)

Worked solution

January 2020 | Question 15 | 5 marks

Family: Optimise a practical model • Variant: Maximum volume from geometric dimensions

Method / working

1. $dV/dx = 16 - 2x - 6x^2$.
 2. Set $dV/dx = 0$: $3x^2 + x - 8 = 0$.
 3. Quadratic formula gives $x = (-1 \pm \sqrt{97})/6$.
 4. The admissible positive root is 1.4748... .
 5. Hence $x = 1.47$ to 3 s.f.
- Final answer: $x = 1.47$

17 A solid, **S**, is made from a hemisphere and a cylinder.

The centre of the circular face of the hemisphere and the centre of the top face of the cylinder are at the same point.

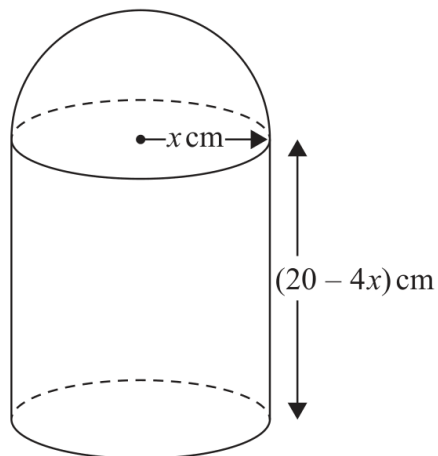


Diagram **NOT**
accurately drawn

The radius of the cylinder and the radius of the hemisphere are both x cm.
The height of the cylinder is $(20 - 4x)$ cm.

The volume of **S** is $V \text{ cm}^3$ where $V = \frac{1}{3} \pi y$

Find the maximum value of y .
Show clear algebraic working.

Worked solution

January 2021 | Question 17 | 5 marks

Family: Optimise a practical model • Variant: Maximum volume from geometric dimensions

Method / working

1. Volume $V = \frac{1}{3}\pi[(\frac{2}{3})x^3 + x^2(20-4x)]$.
 2. Since $V = \frac{1}{3}\pi y$, $y = 60x^2 - 10x^3$.
 3. Differentiate: $dy/dx = 120x - 30x^2$ and set equal to zero.
 4. $x=4$ is the valid stationary value.
 5. Substitution gives $y=320$, the maximum.
- Final answer: $y=320$

- 19 A particle P is moving along a straight line.
The fixed point O lies on this line.

At time t seconds where $t \geq 0$, the displacement, s metres, of P from O is given by

$$s = t^3 + 5t^2 - 8t + 10$$

Find the displacement of P from O when P is instantaneously at rest.

Give your answer in the form $\frac{a}{b}$ where a and b are integers.

..... metres

Worked solution

January 2021 | Question 19 | 5 marks

Family: Apply differentiation to linear kinematics • Variant: Instantaneous rest or direction reversal

Method / working

1. Velocity $v = ds/dt = 3t^2 + 10t - 8$.
2. Set $v = 0$: $(3t - 2)(t + 4) = 0$, so $t = 2/3$ for $t \geq 0$.
3. Substitute $t = 2/3$ into $s = t^3 + 5t^2 - 8t + 10$.
4. Simplify the exact value.
5. Displacement = $194/27$ m.

Final answer: $194/27$ m

- 21** The point A is the only stationary point on the curve with equation $y = kx^2 + \frac{16}{x}$ where k is a constant.

Given that the coordinates of A are $\left(\frac{2}{3}, a\right)$

find the value of a .

Show your working clearly.

$a = \dots\dots\dots$

Worked solution

June 2021 | Question 21 | 5 marks

Family: Find and classify stationary points • Variant: Recover curve parameters from point and gradient data

Method / working

1. Differentiate $y=kx^2+16/x$: $dy/dx=2kx-16/x^2$.
2. At the stationary point $x=2/3$, set $2k(2/3)-16/(2/3)^2=0$, giving $k=27$.
3. Substitute $x=2/3$ into $y=27x^2+16/x$.
4. Evaluate $y=27(4/9)+16/(2/3)=12+24=36$.
5. Therefore $a=36$.

Final answer: $a=36$

23 Curve **C** has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x , in terms of p and m , for which the gradient of **C** is negative.

(Total for Question 23 is 4 marks)

Answer reference | May-Nov 2020 | Question 23

Family: Use the derivative as gradient or rate • Variant: Points where gradient equals a value

23	$\left(\frac{dy}{dx} = 3px^2 - m\right)$		4	M1 for $3px^2$ or $-m$
	$3px^2 - m < 0$ oe			M1 ft dep on M1 for setting up an inequality with their ' $3px^2$ ' - ' m ' must be a two-term expression in the form $apx^2 \pm m$
	$\pm \sqrt{\frac{m}{3p}}$			B1 for both critical values
		$-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$		A1 may be seen as two separate inequalities
Total 4 marks				

Worked solution

May-Nov 2020 | Question 23 | 4 marks

Family: Use the derivative as gradient or rate • Variant: Points where gradient equals a value

Method

$$y = px^3 - mx$$

Differentiate:

$$(dy)/(dx) = 3px^2 - m$$

The gradient is negative when

$$3px^2 - m < 0$$

$$3px^2 < m$$

$$x^2 < (m)/(3p)$$

So

$$-\sqrt{(m)/(3p)} < x < \sqrt{(m)/(3p)}$$

$$\text{Answer: } -\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$$

20 The radius of a right circular cylinder is x cm.

The height of the cylinder is $\left(\frac{800}{\pi x} - x\right)$ cm.

The volume of the cylinder is V cm³

Find the maximum value of V

Give your answer correct to the nearest whole number.



(Total for Question 20 is 5 marks)

Answer reference | January 2023 | Question 20

Family: Optimise a practical model • Variant: Maximum volume from geometric dimensions

20	$(V =) \pi x^2 \left(\frac{800}{\pi x} - x \right)$ or $800x - \pi x^3$ oe		5	M1 for volume of cylinder (in terms of one variable, e.g. x or r)
	$\left(\frac{dV}{dx} = \right) 800 - 3\pi x^2$			M1ft for differentiating an expression in one variable to find 800 or $\pm 3\pi x^2$ (must come from a cubic in the form $800x \pm \pi x^3$ or $800x \pm ax^3$ or $bx \pm \pi x^3$ where $a \neq 0$ and $b \neq 0$)
	" $800 - 3\pi x^2 = 0$ "			M1ft dep on previous M1 for equating their $\frac{dV}{dx}$ to zero (must be a quadratic in the form $800 \pm ax^2$ or $b \pm 3\pi x^2$ where $a \neq 0$ and $b \neq 0$)
	$(x =) \sqrt{\frac{800}{3\pi}} (= \sqrt{84.8(8263...)})$ or $\frac{\sqrt{9600\pi}}{6\pi}$ or $9.2(13177...)$			A1 for a correct value of x Allow use of quadratic formula
	Award marks within the range from correct working	4914		A1 accept 4910 – 4914
				Total 5 marks

Worked solution

January 2023 | Question 20 | 5 marks

Family: Optimise a practical model • Variant: Maximum volume from geometric dimensions

Official final mark-scheme pages are embedded immediately after this question.

22 Curve **C** has equation $y = x^3 - 16x + 7$

At two points on **C**, the gradient is 11

The tangents to **C** at these two points have equations of the form $y = ax + b$

Work out the two possible values of b

Show clear algebraic working.

(Total for Question 22 is 6 marks)

Worked solution

November 2025 | Question 22 | 6 marks

Family: Use the derivative as gradient or rate • Variant: Equation of a calculus tangent

Method

$$y = x^3 - 16x + 7$$

Differentiate:

$$(dy)/(dx) = 3x^2 - 16$$

The gradient of the tangent is 11, so

$$3x^2 - 16 = 11$$

$$3x^2 = 27$$

$$x^2 = 9$$

$$x = 3 \text{ or } x = -3$$

When $x = 3$:

$$y = 3^3 - 16(3) + 7 = 27 - 48 + 7 = -14$$

The tangent has equation $y = 11x + b$, so

$$-14 = 11(3) + b$$

$$b = -47$$

When $x = -3$:

$$y = (-3)^3 - 16(-3) + 7 = -27 + 48 + 7 = 28$$

$$28 = 11(-3) + b$$

$$b = 61$$

Answer: $b = -47$ or $b = 61$